



SIMATS ENGINEERING

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Course Code: CSA 1610

Slot: A

Course Name: Data Warehousing and Data Mining

Course Faculty: Kanimozhi K V

Presentation Title: Query Processing and Business Intelligence Tools



Description:

Query Processing is the process of converting an SQL query into an efficient execution plan and retrieving the required data from the database. It consists of four stages:

- **Parsing** – checks the syntax and validates tables and columns.
- **Translation** – converts the SQL query into an internal representation, such as relational algebra.
- **Query Optimization** – selects the most efficient execution plan using techniques like indexing and join optimization.
- **Execution** – executes the optimized plan and returns the result.

Among these stages, **query optimization** is very important because it improves performance by reducing execution time and system resource usage.

Next, **Business Intelligence (BI) Tools** are software applications that analyze data from data warehouses and present it in a meaningful way. They help organizations make better decisions through dashboards, reports, and data visualization.

Student Signature

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